

Team Name: Books.gov

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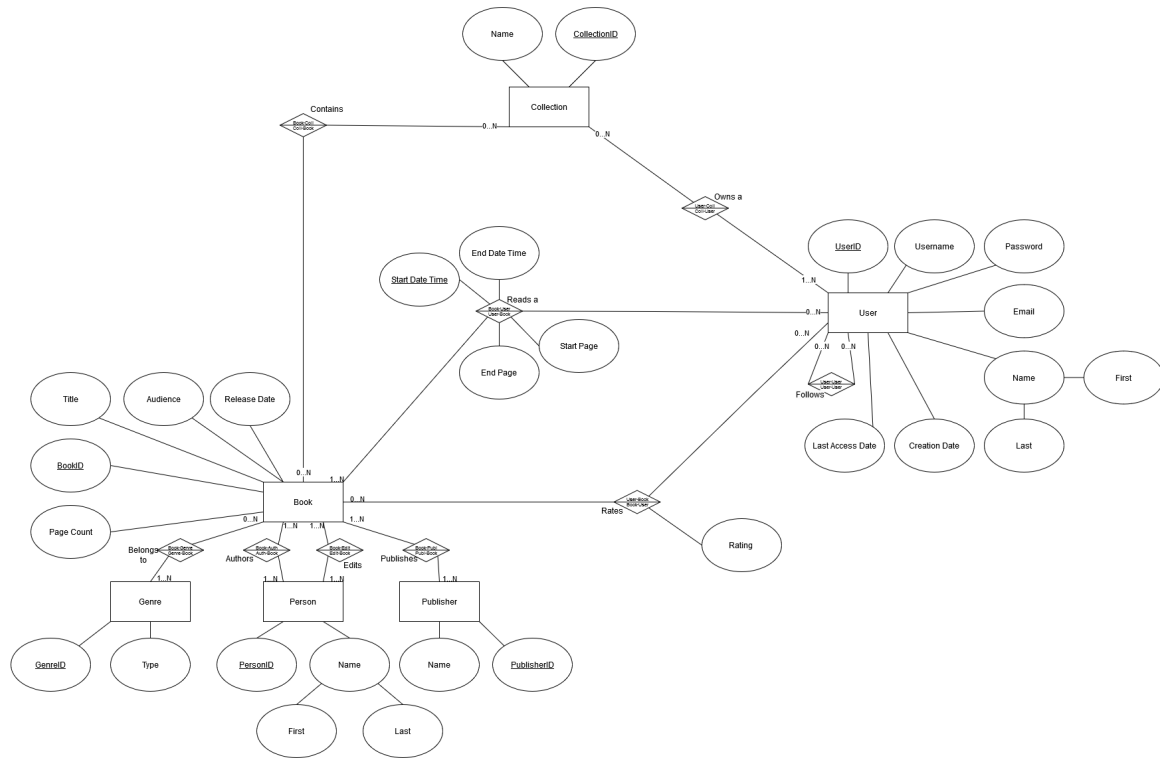
1 Introduction

The database we are designing is meant to manage a user's reading habits and interactions. This database stores essential information about users, books, and interactions between the two. The system stores essential user information such as username, password securely stored, first and last name, email, account creation date, and last access date. This information allows multiple users to use the system effectively. The database also manages book data, including title, intended audience (Kids, Teens, or Adults), release date, length in pages, genres, authors, editors, and publishers. In addition, the database will store unique identifiers and names of authors, editors, and publishers to ensure accurate record-keeping.

Users can take several actions within the system that get stored as well. The system stores sessionID, start and end pages, date, start, and end time whenever a user reads a book. Users can also rate books 1-5, which will subsequently be stored and used to calculate an average rating. Finally, users can create and manage collections of books and follow and be followed by other users.

2 Design

2.1 Conceptual Model



During the development of our EER diagram, we had many different entities and relationships to consider. Initially, we identified a rating and reading session as separate entities that the user creates. However, after further consideration, we realized that those two just describe the relationships between a user and a book. A user reads a book and records the start and end date and page, and a user rates a book, storing the rating. At the beginning of the development of this diagram we had genre as an attribute of a book but due to the wide domain of a genre we opted to put it as its own entity with an id and the genre type as attribute. This will allow a large domain of genres to be input into the database and allows easy access to the many different types. To consolidate our entities we combined the editor and author entities into a single person entity with two different relationships with a book. This streamlines the author and editor data into a single location to store person data and makes it easy to access both relations at once.

2.2 Reduction to Tables

User, Book, Collection, Author, Editor, Publisher, and Genre are all entity types so they all got their own entry in the relational model. For every many-to-many relationship in the EER diagram, an entry in the relational model is needed to link the entities together containing both entities' primary keys.. The entries relating to this are User_Follows, User_Collection, Collection_Book, Book_Author, Book_Editor, Book_Publisher, Book_Genre, User_Book_Rating, User_Book_Session.

User (User_ID, Username, Password, First_Name, Last_Name, Email, Creation_Date, Last_Access_Date)
User_Follows (User_Follower, User_Followed)
User_Collection (User_ID, Collection_ID)
Book (Book_ID, Title, Audience, Length, Release_Date)
Collection (Collection_ID, User_ID, Name)
Collection_Book (Collection_ID, Book_ID)
Author (Author_ID, First_Name, Last_Name)
Book_Author (Book_ID, Author_ID)

Editor (Editor_ID, First_Name, Last_Name)
 Book_Editor (Book_ID, Editor_ID)
 Publisher (Publisher_ID, Name)
 Book_Publisher (Book_ID, Publisher_ID)
 Genre (Genre_ID, Genre_Type)
 Book_Genre (Book_ID, Genre_ID)
 User_Book_Rating (User_ID, Book_ID, Rating)
 User_Book_Session (Start_Time, User_ID, Book_ID, End_Time, Start_Page, End_Page)

2.3 Data Requirements/Constraints

The database for the project will store essential information to support the application requirements. It will manage multiple users by recording details such as username, password (secured in the final phase), first and last name, email, creation date, and last access date. For books, the database will capture the title, audience category (Kids, Teens, Adults), release date, length in pages, genres, authors, editors, and publishers. Additionally, authors, editors, and publishers will be uniquely identifiable by their names. The system will also track user actions, including reading sessions (pages or locations read, start and end times, and date), star ratings for books (1-5, fixed per user per book), creation of named book collections, and user interactions like following and user following.

User:

User_ID, Username, Password, First_Name, Last_Name, Email, Creation_Date, Last_Access_Date must be NOT NULL

UserID, Username must be unique

Creation_Date, Last_Access_Date must be valid date formats

Last_Access_Date must not be before Creation_Date

Users do not need to follow anyone, be followed by anyone, own any collections, rate any books, nor have any reading sessions.

Book:

Book_ID, Title, Audience, Length, Release_Date must be NOT NULL

Book_ID must be unique

Release_Date must be valid dates

Books must have an author, editor, publisher, and genre.

Books do not need to be in a collection, have a rating, or be in any reading sessions.

Collection:

Collection_ID, User_ID, Name must be NOT NULL

Collection_ID, User_ID must be unique

Author: Author_ID, First_Name, Last_Name must be NOT NULL

Author_ID must be unique

Editor:

Editor_ID, First_Name, Last_Name must be NOT NULL

Editor_ID must be unique

Publisher:

Publisher_ID, First_Name, Last_Name must be NOT NULL

Publisher_ID must be unique

Genre:

Genre_ID, Genre_Type must be NOT NULL

Genre_ID, Genre_Type must be unique

User_Follows:

User_Follower, User_Followed must be NOT NULL

User_Collection:

User_ID, Collection_ID must be NOT NULL

Collection_Book:

Collection_ID, Book_ID must be NOT NULL

Book_Author:

Book_ID, Author_ID must be NOT NULL

Book_Editor:
 Book_ID, Editor_ID must be NOT NULL
 Book_Publisher:
 Book_ID, Publisher_ID must be NOT NULL
 Book_Genre:
 Book_ID, Genre_ID must be NOT NULL
 User_Book_Rating:
 User_ID, Book_ID, Rating must be NOT NULL
 Rating must be between 1 and 5 (inclusive)
 User_Book_Session:
 Start_Time, User_ID, Book_ID, End_Time, Start_Page, End_Page must be NOT NULL
 Start_Time, End_Time must be valid date formats
 End_Time must not be before Start_Time
 Start_Page must be non-negative and End_Page must not be less than Start_Page

2.4 Sample instance data

Use this section to include sample of entities for every entity type in your EER diagram. Include also sample of relationships for every relationship type. For example, assume you have an entity type Course in your EER diagram with the attribute types ID and name. A sample of a Course entity can be CSCI320, Principles of Data Management. Include 5 samples for every entity type and relationship type.

Note that your figure will automatically be placed in the most appropriate place for it, given the surrounding text and taking into account other figures or tables that may be close by. You can find out more about adding images to your documents in this help article on [including images on Overleaf](#).

User			
User_ID	Username	Password	ISO numeric Code
1	jdoe123	ma0im43	004
2	JaneSmith22	ALA	248
3	AliceTheReader	ALB	008
4	BobbiBobbi	DZA	012
5	Charlie03	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

3 Implementation

Use this section to describe the overall implementation of your database. Include samples of SQL statements to create the tables (DDL statements) and a description of the ETL process, including examples of the SQL insert statements used to populate each table initially. Include also sample of the SQL insert statements used in your application program to insert new data in the database. Finally, add an appendix of all the SQL statements created in your application during Phase 4 and a description of the indexes created to boost the performance of your application.

Use the table and tabular environments for basic tables — see Table ??, for example. For more information, please see this help article on [tables](#).

4 Data Analysis

4.1 Hypothesis

We expect to see

4.2 Data Preprocessing

Use this section to describe the preprocessing steps you have performed to prepare the data for the analytics. Preprocessing steps may include: data cleaning (e.g., filling missing values, fixing outliers),

formatting the data (e.g., resolving issues like inconsistent abbreviations, multiples date format in the data), combining or splitting fields, add new information (data enrichment). Explain how the data was extracted from the database for the analysis; if you used complex queries or views, or both.

4.3 Data Analytics & Visualization

Use this section to explain the process/techniques used to analyze the data, use data visualization to present the results, and explain them.

4.4 Conclusions

Use this section to explain the conclusions drawn from your data analysis.

5 Lessons Learned

Use this section to describe the issues you faced during the project and how you overcame them. Also, describe what you learned during this effort; this section, like the others, plays a critical component in determining your final grade. **The next subsection is meant to provide you with some help in dealing with figures, tables and references, as these are sometimes hard for folks new to LATEX. Your figures and tables may be distributed all over your paper (not just here), as appropriate for your paper. Please delete the following subsection before you make any submissions!**

5.1 Tables, Figures, and Citations/References

Tables, figures, and references in technical documents need to be presented correctly. As many students are not familiar with using these objects, here is a quick guide extracted from the ACM style guide. First, note that figures in the report must be original, that is, created by the student: please do not cut-and-paste figures from any other paper or report you have read or website. Second, if you do need to include figures, they should be handled as demonstrated here. State that Figure 1 is a simple illustration used in the ACM Style sample document. Never refer to the figure below (or above) because figures may be placed by LATEX at any appropriate location that can change when you recompile your source .tex file. Incidentally, in proper technical writing (for reasons beyond the scope of this discussion), table captions are above the table and figure captions are below the figure. So the truly junk information about flavors is shown in Table 1.

6 Resources

Include in this section the resources you have used in your project beyond the normal code development such as data sets or data analytic tools (i.e. Weka, R).

7 Stuff to eventually delete below

You can make lists with automatic numbering ...

1. Like this,
2. and like this.

...or bullet points ...

- Like this,
- and like this.

7.1 How to write Mathematics

L^AT_EX is great at typesetting mathematics. Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_i^n X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

7.2 How to change the margins and paper size

Usually the template you're using will have the page margins and paper size set correctly for that use-case. For example, if you're using a journal article template provided by the journal publisher, that template will be formatted according to their requirements. In these cases, it's best not to alter the margins directly.

If however you're using a more general template, such as this one, and would like to alter the margins, a common way to do so is via the geometry package. You can find the geometry package loaded in the preamble at the top of this example file, and if you'd like to learn more about how to adjust the settings, please visit this help article on [page size and margins](#).

7.3 How to change the document language and spell check settings

Overleaf supports many different languages, including multiple different languages within one document.

To configure the document language, simply edit the option provided to the babel package in the preamble at the top of this example project. To learn more about the different options, please visit this help article on [international language support](#).

To change the spell check language, simply open the Overleaf menu at the top left of the editor window, scroll down to the spell check setting, and adjust accordingly.

7.4 How to add Citations and a References List

You can simply upload a `.bib` file containing your BibTeX entries, created with a tool such as JabRef. You can then cite entries from it, like this: [\[Gre93\]](#). Just remember to specify a bibliography style, as well as the filename of the `.bib`. You can find a [video tutorial here](#) to learn more about BibTeX.

If you have an [upgraded account](#), you can also import your Mendeley or Zotero library directly as a `.bib` file, via the upload menu in the file-tree.

7.5 Good luck!

We hope you find Overleaf useful, and do take a look at our [help library](#) for more tutorials and user guides! Please also let us know if you have any feedback using the Contact Us link at the bottom of the Overleaf menu — or use the contact form at <https://www.overleaf.com/contact>.

References

- [Gre93] George D. Greenwade. The Comprehensive Tex Archive Network (CTAN). *TUGBoat*, 14(3):342–351, 1993.